AIRBNB PRICING ANALYSIS

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PROJECT REPORT

PREPARED AND PRESENTED BY

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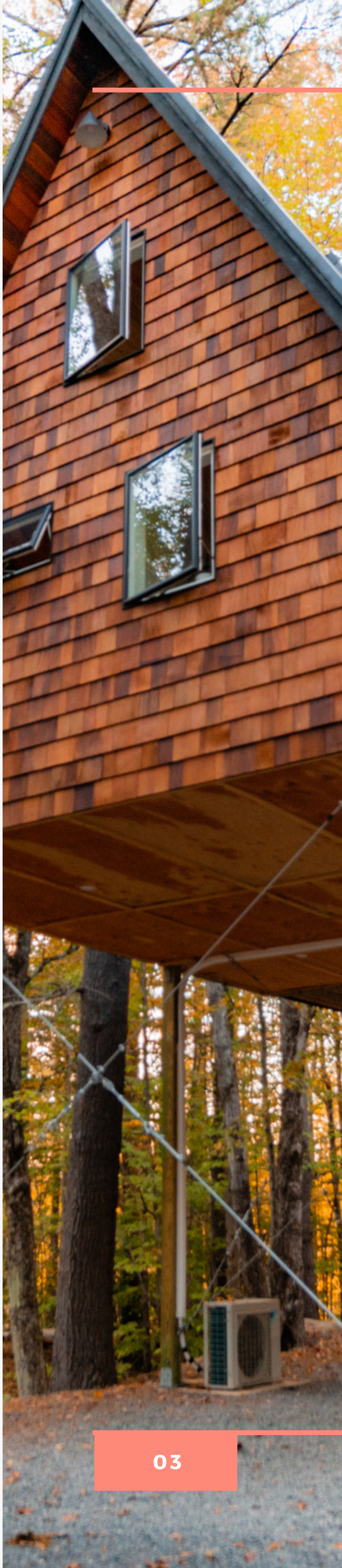
THE OBJECTIVE

AIRBNB PRICING ANALYSIS

The key objective is to identify the key factors that influence Airbnb listing prices in New York City and to develop a predictive pricing model using publicly available data. As it stands, many Airbnb hosts still rely on intuition to set nightly rates. Our project seeks to replace this gut feel approach with a predictive pricing model trained on Airbnb's NYC listings. Our predictive model helps the Airbnb hosts adjust their property locations, features & pricing to maximize their revenue while aligning with guest expectations. This project explores how property features, location, and host behaviour contribute to nightly rates and builds a model that provides insights and guidance for both individual hosts and the Airbnb platform.

APPROACH SUMMARY

- Started by cleaning the data—removed unnecessary details (like listing names and IDs) and filled in missing values.
- Calculated how many days had passed since a listing's last guest review to understand recent activity.
- Flagged listings where hosts managed multiple properties, to spot more "professional" setups.
- Grouped listings based on how many reviews they had and how long guests were required to stay.
- Combined neighbourhood and room type into a single feature to better capture pricing patterns by location.
- Translated descriptive info (like room type or neighbourhood) into a format the model could understand.
- Used a predictive model to focus on the features most strongly linked to price.
- Using a log price function, we created a predictive tool for users to input values related to features such as neighbourhood group, room type, minimum nights, reviews per month, host listing counts to give a prediction of what the per night charge for an airbnb listing should be.



ANALYSIS

To explore the factors that influence Airbnb listing prices in New York City, we developed a **predictive pricing model** that achieved reasonable accuracy, capturing over **54% of the variation** in nightly prices. We trained a **LassoCV model with 5-fold cross-validation**. Lasso was selected for its interpretability, embedded feature selection, and ability to handle high-dimensional data. We pre-processed data by filtering invalid listings (price <= 0) ,Engineered key features: days_since_last_review, region_room, binned minimum_nights & reviews, One-hot encoded categorical variables (neighbourhood, room type, etc.), Log-transformed price to handle skew, did an 80/20 train-test split, and scaled numeric variables using a StandardScaler function. We then modelled the data using a LassoCV model with 5-fold cross-validation. Lasso was selected for its interpretability, embedded feature selection, and ability to handle high-dimensional data.

The key results we got from our model: R^2 (test set): 0.543, RMSE: $185.96 (log price transformed back), and an Optimal alpha: 0.00023. This means that the model captures over 54% of the variance in price using only listing-level features. The RMSE of ~$186 suggests moderate predictive precision, competitive with Airbnb's internal host estimates. Our model also identifies the top price drivers (based on Lasso coefficients). Predictions were generally more accurate for listings below $400/night, which make up the bulk of the market. Based on our analysis, the most important factor in determining price is room type. Listings categorized as **"Shared room" were priced approximately $105 lower per night than "Entire home/apt" listings, while "Private room" listings were about $65 lower.** This pattern reflects guests' strong preference and willingness to pay for full privacy. Location was another major driver. Listings located in Manhattan were priced roughly $95 higher than those in other boroughs, with premium neighbourhoods like Tribeca, Brooklyn Heights, and Washington Heights consistently associated with higher nightly rates. These areas likely benefit from a combination of reputation, centrality, and proximity to major attractions. We also split the 5 boroughs of NYC into a "neighbourhood group" type to account for any influences from the boroughs. We expected price to be more influenced by whether a listing was in Manhattan rather than Brooklyn, but the results seem to indicate otherwise. It seems that the neighbourhood a listing is in has more influence on the price than what borough it is located in. This can be attributed to gentrification, certain neighbourhoods having more demand than others. In addition to property attributes, we also observed that listings managed by hosts with multiple properties were more likely to have frequent reviews, suggesting professional management and potentially more refined pricing strategies. Furthermore, recent review activity, measured by how many days had passed since the last guest review, correlated with higher prices, indicating that more active and in-demand listings tend to command better rates.

RECOMMENDATION

Based on our findings, we propose several actionable recommendations aimed at improving pricing accuracy, listing competitiveness, and host decision-making on the Airbnb platform.

For Individual Hosts: New and existing hosts should move beyond simply benchmarking prices against neighbourhood averages. Our analysis reveals that factors such as room type, review volume, host listing behaviour, and minimum-stay requirements play critical roles in determining price. For example, listings with recent reviews and shorter minimum-stay policies tend to command higher nightly rates. We recommend that hosts evaluate these dimensions when setting prices. Multi-listing hosts, in particular, may benefit from employing automated pricing tools, A/B testing, or third-party analytics dashboards to experiment with dynamic pricing strategies and monitor occupancy versus revenue trade-offs. These tools can help maintain competitive positioning while maximizing income.

For Airbnb as a Platform: Our model reveals that structured listing attributes provide strong signals for price prediction, suggesting that Airbnb could enhance its pricing recommendation engine by incorporating engineered features such as region_room, reviews_bin, or host portfolio size. By identifying patterns in high-performing listings, Airbnb can proactively suggest pricing guidelines tailored to listing characteristics and host type. Additionally, Airbnb could implement onboarding nudges for new hosts — for instance, recommending shorter minimum stays or encouraging early reviews — based on what our analysis found to be significant price influencers.

Shared Benefits: By making pricing decisions more data-informed and responsive to evolving market signals, both hosts and the platform can benefit. Hosts gain greater confidence and control in setting competitive, optimized rates, while Airbnb improves marketplace efficiency and trust. Enhanced pricing transparency can lead to improved guest experiences, reduced vacancy rates, and stronger platform loyalty. In a competitive urban market like NYC, these marginal gains translate to significant cumulative impact. Future iterations of this work could expand the model to include textual sentiment from reviews, guest ratings, and seasonal pricing dynamics, allowing for even more refined pricing guidance.

CONCLUSION

These findings suggest that both listing characteristics and host behaviours significantly shape Airbnb pricing across New York City. Properties offering full-unit privacy and located in central, well-known neighbourhoods have a clear advantage in commanding higher prices. Room type and location together form a strong foundation for price differentiation, confirming the importance of visibility, accessibility, and perceived exclusivity in short-term rentals. Host strategy also appears to play a role.

Multi-listing hosts may be more deliberate in how they set prices and manage turnover, while guest activity, measured through recent reviews, seems to reflect and reinforce listing popularity. In contrast, longer minimum-stay requirements are often linked with discounted nightly rates, indicating a trade-off between occupancy and per-night earnings. The model's ability to explain over half of the variance in price using only observable listing data provides strong justification for its use in pricing support tools. While more granular data or nonlinear models may improve performance further, our current framework demonstrates that interpretable, actionable insights can be delivered through careful preprocessing and linear regularization.

APPENDIX

Code: (See attached FinalGroupProjectFinalModel.ipynb for full code) We implemented a fully aligned prediction function replicating the training pipeline. For the following listing: { 'neighbourhood_group': 'Manhattan', 'neighbourhood': 'Harlem', 'room_type': 'Private room', 'minimum_nights': 3, 'number_of_reviews': 20, 'reviews_per_month': 1.5, 'calculated_host_listings_count': 2, 'availability_365': 180, 'latitude': 40.8090, 'longitude': -73.9440, }

The model predicted: $143.40 per night The actual listing price was approximately $132.06, leading to an absolute error of ~$11.34 (≈8.6%). This is well within the model's expected margin of error and demonstrates acceptable accuracy for real-world use.

Data: See AB_NYC_2019.csv

Top 10 Coefficients Table

room_type_Shared room	1.049478
room_type_Private room	0.677344
neighbourhood_group_Manhattan	0.495641
neighbourhood_Tribeca	0.491630
neighbourhood_Washington Heights	0.440851
neighbourhood_Inwood	0.400587
neighbourhood_group_Staten Island	0.324726
min_nights_bin_8-30	0.280644
neighbourhood_Williamsburg	0.279640
neighbourhood_Brooklyn Heights	0.274501

VISUALIZATIONS